

CHE 321

1. Course number and name

CHE 321. Chemical Reaction Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Vikas Berry, Meenesh Singh

4. Textbook, title, author, publisher, year

"Elements of Chemical Reaction Engineering",
Fifth, Fourth or Third Edition, by Fogler, H. Scott

5. Specific course information

- a. Brief description of the content of the course (catalog description)
Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors. Non-isothermal reactors.
- b. Prerequisites or co-requisites
CHE 210, CHE 301, MATH 220.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

- a. Specific outcomes of instruction.
 1. Compare batch reactors, continuous stirred tank reactors (CSTR), and plug flow reactors (PFR).
 2. Derive, solve the mass balance for isothermal reaction in a CSTR, PFR, or batch reactor.
 3. Calculate the equilibrium conversion in reactor type for isothermal or adiabatic conditions.
 4. Solve the governing mass and energy balances for a CSTR, PFR, packed bed, or batch reactor.
 5. Estimate reaction orders, reaction rate constants, and Arrhenius parameters from experimental kinetic data.
 6. Solve the governing mass balances for multiple reactions in a CSTR, PFR, or batch reactor
 7. Suggest which type of continuous reactor (CSTR or PFR) is advantageous for maximizing product selectivity for series or parallel reactions
 8. Derive a Langmuir-Hinshelwood kinetic rate law
 9. Incorporate a mass transfer coefficient into a reaction rate expression to account for external mass transfer limitations
 10. Calculate the effectiveness factor for a catalyst particle given physical properties of the particle, the reaction rate constant, and the diffusion coefficient for the reactant
 11. Calculate the residence time distribution from a tracer study.
 12. Draw similarity between electron transport through nanoparticles and chemical reactions.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. **Brief list of topics to be covered**

Reaction fundamentals

Reactor Types, General Mole Balances, Conversion, Reactor Sizing, Stoichiometric Tables

Isothermal Reactor Design

SOFTWARE - Polymath and MaTLAB for Reactor Design

Pressure Drop in Reactors

Unsteady State Operation, Rate Analysis

Multiple Reactions

General Energy Balance & CSTR, PFR Energy Balance, Batch Energy Balance

Equilibrium Conversion

Non-Isothermal Reactor Example

Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions

Summary of Non-Isothermal Reactors

Catalytic Reactions, Langmuir-Hinshelwood Rate Laws

Langmuir-Hinshelwood Rate Laws, Heterogeneous Reactors, Mass Transfer

External and Internal Mass Transfer Limitations

Effectiveness Factor and Falsified Kinetics

Residence Time Distribution

Zero-Parameter Flow Models, Non-Ideal Flow and combination of ideal reactors